

B.Tech. Degree III Semester Regular/Supplementary Examination November 2023

CE/CS/EC/EE/IT/ME/SE 19-200-0301 LINEAR ALGEBRA AND TRANSFORM TECHNIQUES
(2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

Course Outcomes

On successful completion of the course, the students will be able to:

CO1: Solve linear system of equations and to determine eigen values and eigen vectors of a matrix.

CO2: Understand the concept of vector space and subspace.

CO3: Determine Fourier series expansion of functions and transform.

CO4: Solve linear differential equation and integral equation using Laplace transform.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create

PO – Programme Outcome

PART A

(Answer ALL questions)

	(8 × 3 = 24)	Marks	BL	CO	PO
I. (a) Applying Cayley–Hamilton theorem find A^2 and A^{-1} if $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$.	3	3	L3	1	2
(b) Reduce the quadratic form $3x^2 - 3y^2 + 8xy$ in two variables into Canonical form.	3	3	L2	1	1
(c) Whether the vectors $V_1 = (2, 2, 0)$, $V_2 = (3, 0, 2)$, $V_3 = (2, -2, 2)$ in R^3 are linearly independent. Whether $\{V_1, V_2, V_3\}$ form a basis of R^3 . Why?	3	3	L2	2	2
(d) Define: (i) A Linear transformation from a vector space to another vector space. (ii) Kernel of a Linear transformation.	3	3	L1	2	1
(e) Find the Fourier sine transform of $f(x) = e^{-2x}$	3	3	L2	3	2
(f) Define period of function and find the period of $f(x) = \sin \pi x + \cos \pi x + \tan \pi x$.	3	3	L1 L2	3	2
(g) Find the inverse Laplace transform of $\log \left(\frac{s^2 + 1}{s(s+1)} \right)$.	3	3	L4	4	3
(h) Evaluate $\int_0^{\infty} \frac{e^{-t} \sin t}{t} dt$.	3	3	L5	4	3

PART B

	(4 × 12 = 48)	Marks	BL	CO	PO
II. (a) Reducing to the normal form find the Rank of the matrix $A = \begin{bmatrix} 0 & 1 & 2 & -2 \\ 4 & 0 & 2 & 6 \\ 2 & 1 & 3 & 1 \end{bmatrix}$	6	6	L3	1	2
(b) The matrix $A = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 2 \end{bmatrix}$ then find Eigen values of (i) A (ii) A^T (iii) A^{-1} (iv) Adjoint A	6	6	L2 L4	1	1

OR

(P.T.O.)

- III. (a) Test for consistency and if consistent solve the following system of equations:

$$\begin{aligned}x + 3y + z &= 4 \\2x + 3y + 2z &= 5 \\3x - 5y + 5z &= 2 \\3x + 9y - z &= 4\end{aligned}$$

- (b) Diagonalise the matrix $A = \begin{bmatrix} -1 & 3 \\ -2 & 4 \end{bmatrix}$ and hence find A^4 .

6	L2, L4	1	3
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- IV. (a) Find the matrix of linear transformation with respect to the basis

$$\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\} \text{ of } T: R^3 \rightarrow R^3 \text{ such that } T \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2x - y \\ 0 \\ y + z \end{bmatrix}$$

5	L4	2	2
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- (b) Let $v_1 = (3, 0, 4)$, $v_2 = (-1, 0, 7)$, $v_3 = (2, 9, 11)$ are vectors in R^3 equipped with the standard inner product. Using Gram-Schmidt orthogonalization process obtain an orthogonal basis and ortho normal basis for R^3 .

7	L3	2	2
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OR

- V. (a) Determine whether the mapping $T: R^2 \rightarrow R^3$ defined by $T(x, y) = (x, y, 0)$ is a linear transformation or not.

4	L3	2	2
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- (b) Let W be the set of all points in R^3 satisfying the equation $ax + by + cz = 0$. Show that W is a subspace of R^3 .

4	L4	2	2
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- (c) Let $V(R)$ be a vector space of polynomials with inner product defined by $\langle f, g \rangle = \int_0^1 f(x) g(x) dx$. If $f(x) = x^2 + x - 4$, $g(x) = x - 1$.

4	L3	2	2
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Find $\langle f, g \rangle$ and $\|g\|$.

- VI. (a) Find the Fourier series for the function $f(x) = x + x^2$ in the interval $-\pi < x < \pi$.

6	L4	3	2
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- (b) Using Fourier integral prove that $\int_0^\infty \frac{\cos x \lambda}{1 + \lambda^2} d\lambda = \frac{\pi}{2} e^{-x}, x \geq 0$.

6	L4	3	2
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OR

- VII. (a) Find the half range sine series for $f(x) = 2 - x$ for $0 < x < 2$ and hence deduce that $1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots = \frac{\pi}{4}$.

6	L4	3	2
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- (b) Find the Fourier transform of $f(x) = e^{-\frac{x^2}{2}}$.

6	L4	3	2
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- VIII. (a) Define unit step function and find its Laplace transform.

4	L4, L3	4	2
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- (b) Applying Convolution theorem find the inverse Laplace transform of

4	L3	4	2
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$$\frac{1}{s(s^2 + 4)}$$

- (c) Evaluate $\int_{-\infty}^{\infty} e^{-x^2} dx$.

4	L5	4	3
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OR

- IX. (a) State and prove second shifting theorem.

4	L1, L3	4	2
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- (b) Find the Laplace transform of $\frac{\cos at - \cos bt}{t}$.

4	L3	4	2
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- (c) Solve using Laplace transform $y'' + y = t$, $y(0) = 1$, $y'(0) = 0$.

4	L3	4	3
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Bloom's Taxonomy Levels

L1 = 7%, L2 = 19%, L3 = 36%, L4 = 28%, L5 = 10%.
